



Danmark Tekniske Universitet
TBA

DOCTORADO EN ECOLOGIA TERRESTRE

Reinforcement Learning-based multimodal traffic signal control considering pedestrians and cyclists in real scenarios

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Però tu sabràs el que fas, jo vindria a practicar el noble art d'anar endavant.
Si et sents intrús no parlis gaire i ja veuràs, ningú adverteix l'engany;
no et demanen un carnet, no hi ha escrit cap reglament, s'equivoca el més expert.
Donem conversa, som amables, i no se sap si malgrat l'esforç anem
cap al plaer o al dolor,
a la llum o a la foscor,
a un gran banquet o a la més cruel inanició.
Tots ens movem, no pots quedar-te encallat eternament!
Avança, vianant, per les llambordes i l'asfalt!

MANEL – *Avança, vianant*

To the memory of Miquel Àngel Llauger i Lull
1926 – 2009

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All human endeavours are collaborative, we all grow when surrounded by the ones we love and admire. This thesis carries, directly or indirectly, the imprint of many people that have shared with me parts of this journey. Academically, I am extremely grateful to Roberto and Miguel, for their help and advice throughout these years. At CREAF-UAB, I would like to thank the whole team for such a great atmosphere, and in particular Pep Piñol, Raúl García-Valdés, Jordi Martínez-Vilalta, Miquel Ninyerola, Llorenç Sáez and Mariona Ferrandiz for offering me the opportunity to participate in teaching activities, and Yolanda Melero and Alberto Navarro for collaborating in these duties.

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Aside from the colleagues who have helped me along the way, I think it's necessary to acknowledge the work of many people from which I have benefited. Publicly available ecological datasets such as the ones used in chapter 4 are the result of the effort of countless technicians and volunteers, alongside researchers. Likewise, the R community and, in general, the open software movement, have shown me that it is completely possible to develop robust, reproducible science without resorting to proprietary software at all. I am, all ecologists are, in debt with the communities that provide these basic tools and data for our work.

I am purposely keeping these acknowledgements in the academic side of things. At a personal level, the imprint is even more important, and I could not have carried out this thesis without the love and support of those close to me. Thanks to all of you.

Kongens Lyngby, Danmark
July 2026

ABSTRACT

Organisms survive, thrive and reproduce by interacting with individuals of their own and of other species. Biotic interactions are extremely diverse in type, magnitude, or spatiotemporal scale, and give rise to ecological networks with complex topologies and dynamics. Such networks of ecological interactions have been shown to possess non-random structural properties that enhance their resilience and robustness to perturbations, and thus are key elements for understanding the response of species to external forcing such as environmental change or habitat loss.

Despite the importance of interaction networks in studies of ecological communities, and due in part to their sheer variability, ecological interactions are notoriously difficult to document and quantify in a comprehensive fashion. Therefore, historically, studies of ecological networks have focused on the most easily observable types of interactions, those between predators and their prey, or more generally speaking, between consumers and resources. In the last decades, studies of mutualistic networks have also risen to prominence and have demonstrated, for example, that food webs and mutualistic networks have markedly different topologies and this has both ecological and evolutionary consequences for the species involved. One of the main challenges of contemporary community ecology is to expand our understanding of networks of a single interaction type to a more realistic view of ecological communities, by considering how different interactions mutually influence community structure and functioning. In order to tackle this challenge, a first step is to lay down overarching theoretical hypotheses about such complex networks.

In this thesis I approach this general objective and analyse a series of fundamental questions about ecological networks. After a general introduction, first I synthesise current methodologies for developing theoretical network models. I find that three main conceptual approaches have been used, and discuss their relative strengths, weaknesses, and potential uses. Second, I study whether species persistence in model communities is influenced by the frequency and distribution of the different interaction types. The prevalence of positive interactions within a community is shown to be key for species-poor communities, whereas in more speciose communities, different combinations of interactions can occur without affecting species persistence in a significant way. Furthermore, networks with randomly distributed interactions show less species persistence than structured networks. If community structure is important for species persistence, it follows that other community-level patterns should also be affected by it. In the fourth chapter, I focus on Species Abundance Distributions (SADs), one of the most studied patterns in community ecology, and ask whether their shape varies in a consistent way for the different trophic guilds of a community. I compare theoretical expectations with SADs from empirical datasets, and find that SADs of plant communities are significantly less even and more skewed than SADs from mammal ones. Among mammal trophic guilds, there are no significant differences in the evenness or skewness of their SADs. These first chapters deal with the structure and dynamics of closed communities, aiming to establish baseline hypotheses. In the fifth chapter, I incorporate another degree of complexity, namely the spatial perspective. Specifically, I analyse how interaction effects are propagated in space, such that interactions occurring in a local community may influence other communities connected to it by means of dispersing or foraging

individuals. Given the novelty of this analysis and the long tradition of food web models, in this chapter I focus on trophic communities as a simplified model system. I find that the distribution of net effects of a species over another across the metacommunity is significantly different if the local communities are connected by dispersal, foraging, or a mixture of both. In the sixth chapter, I tackle the long-standing question of the variability of species interactions across environmental gradients. For approaching this question, I recover the distinction, originally proposed by G.E. Hutchinson, about scenopoetic and bionomic environmental factors, i.e. non-resource and resource factors. By recognizing that these two types of environmental factors have different effects on species fitness and on the importance of species' pairwise interactions, I analyse the prevalence of positive and negative interactions in model communities across a two-dimensional environmental gradient with one resource and one non-resource factor. I find that, according to the expectations, positive interactions respond to the non-resource factor, whereas negative interactions vary across the two axis of the gradient, with consequences for average persistence time and species diversity across the combined gradient.

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1.1 Motivation

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1.2 Problem statement

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1.3 Research objectives

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1.4 Scope and limitations



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2.1 Traffic signal control

(?) Where adjacent junctions in a network of urban road are less than a mile apart and are controlled by traffic signals, major benefits can be obtained by installing an urban traffic control (UTC) system to coordinate the operation of the signals. Such systems use a central computer to control the signal sequences and to monitor their operation; the signals and computer are usually connected by voice grade data transmission lines. The method of controlling the signal sequences can have a significant effect on levels of urban congestion, on fuel economy, and on exhaust emissions. This is the main topic of this paper. The first UTC systems came into operation in the mid-1960's, and since then their use has been growing rapidly.

2.2 Modelling the behaviour of pedestrian and cyclists

2.3 Reinforcement learning in traffic signal control

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3.1 Simulation environment

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3.2 Scenario design

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3.3 Reinforcement learning framework

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3.4 Definitions and architecture

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4.1 Evaluation metrics

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4.2 Results

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5.1 Interpretation of results

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5.2 Implications for urban mobility

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5.3 Limitations of the study

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6.1 Conclusions

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6.2 Future work

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Appendix A

Appendix 2.1: Chapter 2 - Case study model parameterizations and procedure

Expanded Food Web

The model is of the form:

$$\frac{dN_x}{dt} = r_x N_x - m_x N_x^2 + \sum_{y \in S, y \neq x} a_{xy} N_x N_y \quad (\text{A.1})$$

where S is the set of species. Non-trophic interactions (NTI) may affect the growth rate r_x , mortality rate m_x or interaction strength a_{xy} terms. The ones included in our model are, respectively:

growth rates

$$r_{CM} = \frac{r_{CM}^{NTI} N_{PL} + r_{CM}^0 N_{PL}^0}{N_{PL} + N_{PL}^0} \quad (\text{A.2})$$

$$r_{DP} = \frac{r_{DP}^{NTI} N_{HM} + r_{DP}^0 N_{HM}^0}{N_{HM} + N_{HM}^0} \quad (\text{A.3})$$

$$r_{PL} = \frac{r_{PL}^{NTI} (N_{CM} + N_{HM} + N_{PLe}) + r_{PL}^0 (N_{CM}^0 + N_{HM}^0 + N_{PLe}^0)}{(N_{CM} + N_{HM} + N_{PLe}) + (N_{CM}^0 + N_{HM}^0 + N_{PLe}^0)} \quad (\text{A.4})$$

$$r_{PLe} = \frac{r_{PLe}^{NTI} N_{PL} + r_{PLe}^0 N_{PL}^0}{N_{PL} + N_{PL}^0} \quad (\text{A.5})$$

mortality rates

$$m_{HM} = \frac{m_{HM}^{NTI} N_{SV} + m_{HM}^0 N_{SV}^0}{N_{SV} + N_{SV}^0} \quad (\text{A.6})$$

interaction terms

$$a_{PL,DP} = \frac{a_{PL,DP}^{NTI} N_{HM} + a_{PL,DP}^0 N_{HM}^0}{N_{HM} + N_{HM}^0} \quad (\text{A.7})$$

$$a_{DP,PL} = \frac{a_{DP,PL}^{NTI} N_{HM} + a_{DP,PL}^0 N_{HM}^0}{N_{HM} + N_{HM}^0} \quad (\text{A.8})$$

We performed two sets of simulations, with and without non-trophic interactions (NTI henceforth). In the simulation without NTI, we assumed that *Podarcis lilfordi* (PL) consumed seeds of *Helicodiceros muscivorus* (HM), *Pistacia lentiscus* (Ple) and *Chritum maritimum* (CM); these interactions were modelled as mutualisms in the NTI simulation, with the presence of PL individuals increasing the growth rate of each associated plant species. Each simulation was replicated 100000 times for 2500 timesteps. In the main text we report the aggregated results of the 100000 replicates. Each parameter was assigned a minimum and maximum value, and in each replicate parameter values were taken randomly from these intervals.

Parameter ranges for the simulation without NTIs:

Growth and mortality rates:

$$\begin{aligned} r_{FT} &= [-0.01, -0.001] \\ r_{PL} &= [0.01, 0.1] \\ r_{DP} &= [0.4, 0.6] \\ r_{SV} &= [0.05, 0.15] \\ r_{HM} &= [0.15, 0.25] \\ r_{PLe} &= [0.05, 0.15] \\ r_{CM} &= [0.15, 0.25] \\ m_{FT} &= [0.0001, 0.0015] \\ m_{PL} &= [0.0001, 0.0015] \\ m_{DP} &= [0.0005, 0.0015] \\ m_{SV} &= [0.0005, 0.0015] \\ m_{HM} &= [0.0005, 0.0015] \\ m_{PLe} &= [0.0002, 0.0004] \\ m_{CM} &= [0.00005, 0.00015] \end{aligned}$$

Interaction coefficients $\neq 0$:

$$\begin{aligned}
a_{FT,PL} &= [10^{-6}, 10^{-4}] \\
a_{PL,FT} &= [-10^{-2}, -10^{-4}] \\
a_{PL,DP} &= [10^{-4}, 10^{-2}] \\
a_{PL,HM} &= [10^{-6}, 10^{-4}] \\
a_{PL,PLe} &= [10^{-7}, 10^{-5}] \\
a_{PL,CM} &= [10^{-7}, 10^{-5}] \\
a_{DP,PL} &= [-10^{-5}, -10^{-7}] \\
a_{DP,HM} &= [10^{-6}, 10^{-4}] \\
a_{HM,PL} &= [-10^{-6}, -10^{-8}] \\
a_{HM,DP} &= [-10^{-6}, -10^{-8}] \\
a_{PLe,PL} &= [-10^{-6}, -10^{-8}] \\
a_{CM,PL} &= [-10^{-6}, -10^{-8}]
\end{aligned}$$

Initial abundances:

$$\begin{aligned}
N_{FT} &= 8 \\
N_{PL} &= 5000 \\
N_{DP} &= 200 \\
N_{SV} &= 5000 \\
N_{HM} &= 5000 \\
N_{PLe} &= 200 \\
N_{CM} &= 5000
\end{aligned}$$

Parameter ranges for the simulation with NTIs. Parameters without superscript are not affected by NTIs. Parameters with superscript 0 indicate values in the absence of NTI, i.e. when one of the interacting species is not present. Parameters with superscript NTI indicate the maximum value that the parameter can reach with NTI:

$$\begin{aligned}
r_{FT} &= [-0.01, -0.001] \\
r_{PL}^0 &= [0.01, 0.1] \\
r_{PL}^{NTI} &= [0.15, 0.25] \\
r_{DP}^0 &= [0.65, 0.75] \\
r_{DP}^{NTI} &= [0.4, 0.6] \\
r_{SV} &= [0.05, 0.15] \\
r_{HM}^0 &= [0.15, 0.25] \\
r_{HM}^{NTI} &= [0.45, 0.55] \\
r_{PLe}^0 &= [0.05, 0.15] \\
r_{PLe}^{NTI} &= [0.15, 0.25] \\
r_{CM}^0 &= [0.15, 0.25] \\
r_{CM}^N TI &= [0.35, 0.45] \\
m_{FT} &= [0.0001, 0.0015] \\
m_{PL} &= [0.0001, 0.0015] \\
m_{DP} &= [0.0005, 0.0015] \\
m_{SV} &= [0.0005, 0.0015] \\
m_{HM}^0 &= [0.0005, 0.0015] \\
m_{HM}^N TI &= [0.00005, 0.00015] \\
m_{PLe} &= [0.0002, 0.0004] \\
m_{CM} &= [0.00005, 0.00015]
\end{aligned}$$

In this simulation, only the interactions PL-DP and FT-PL are considered trophic. Therefore, we report only a values for these. The FT-PL interaction is not affected by any third species, but the PL-DP interaction is mediated by the presence of HM plants: a higher abundance of HM flowers increases the probabilities that an interaction takes place, therefore increasing its net outcome.

$$\begin{aligned}
a_{FT,PL} &= [10^{-6}, 10^{-4}] \\
a_{PL,FT} &= [-10^{-2}, -10^{-4}] \\
a_{PL,DP}^0 &= [10^{-7}, 10^{-5}] \\
a_{PL,DP}^N TI &= [10^{-4}, 10^{-2}] \\
a_{DP,PL}^0 &= [-10^{-5}, -10^{-7}] \\
a_{DP,PL}^N TI &= [-10^{-2}, -10^{-4}]
\end{aligned}$$

The N_0 parameters in previous equations represent a typical average abundance of the non-trophic interactor. These values were taken, when possible, from ???. Diptera densities were approximated based on ??:

$$\begin{aligned}
N_{PL}^0 &= 2189 \\
N_{DP}^0 &= 150 \\
N_{SV}^0 &= 3000 \\
N_{HM}^0 &= 7187 \\
N_{PLe}^0 &= 187 \\
N_{CM}^0 &= 10000
\end{aligned}$$

Equal Footing Network

The model is of the form:

$$\frac{dN_x}{dt} = r_x N_x \tag{A.9}$$

where

$$r_x = r_x^0 + \sum_{y \in S, y \neq x} a_{xy} N_y - \left(\beta_x + c_x \sum_{y \in S, y \neq x} a_{xy} N_y \right) N_x \tag{A.10}$$

As with the Expanded Food Web model, we constrained each free parameter to a given range and simulated 100000 times the system for 2500 time steps, assigning a random value to each parameter within its range. Here we varied the strength of the antagonistic and facilitative interactions (commensalistic and mutualistic) and checked the stability of the resulting network by means of a local stability analysis.

$$\begin{aligned}
r_{FT} &= [-0.01, -0.001] \\
r_{PL} &= [0.01, 0.1] \\
r_{DP} &= [0.4, 0.6] \\
r_{SV} &= [0.05, 0.15] \\
r_{HM} &= [0.15, 0.25] \\
r_{PLe} &= [0.05, 0.15] \\
r_{CM} &= [0.15, 0.25] \\
\beta_x &= [10^{-5}, 10^{-4}] \forall x \\
c_x &= 10^{-3} \forall x
\end{aligned}$$

Interaction strengths

1. Weak interactions

$$\begin{aligned}
a_{FT,PL} &= [10^{-7}, 10^{-5}] \\
a_{PL,FT} &= [-10^{-5}, -10^{-7}] \\
a_{PL,DP} &= [10^{-7}, 10^{-5}] \\
a_{PL,HM} &= [10^{-7}, 10^{-5}] \\
a_{PL,PLe} &= [10^{-7}, 10^{-5}] \\
a_{PL,CM} &= [10^{-7}, 10^{-5}] \\
a_{DP,PL} &= [-10^{-5}, -10^{-7}] \\
a_{DP,HM} &= [10^{-7}, 10^{-5}] \\
a_{HM,PL} &= [10^{-7}, 10^{-5}] \\
a_{HM,DP} &= [10^{-7}, 10^{-5}] \\
a_{PLe,PL} &= [10^{-7}, -10^{-5}] \\
a_{CM,PL} &= [10^{-7}, -10^{-5}]
\end{aligned}$$

2. Strong antagonisms

$$\begin{aligned}
a_{FT,PL} &= [10^{-4}, 10^{-2}] \\
a_{PL,FT} &= [-10^{-2}, -10^{-4}] \\
a_{PL,DP} &= [10^{-4}, 10^{-2}] \\
a_{PL,HM} &= [10^{-7}, 10^{-5}] \\
a_{PL,PLe} &= [10^{-7}, 10^{-5}] \\
a_{PL,CM} &= [10^{-7}, 10^{-5}] \\
a_{DP,PL} &= [-10^{-2}, -10^{-4}] \\
a_{DP,HM} &= [10^{-7}, 10^{-5}] \\
a_{HM,PL} &= [10^{-7}, 10^{-5}] \\
a_{HM,DP} &= [10^{-7}, 10^{-5}] \\
a_{PLe,PL} &= [10^{-7}, -10^{-5}] \\
a_{CM,PL} &= [10^{-7}, -10^{-5}]
\end{aligned}$$

3. Strong facilitation

$$\begin{aligned}
a_{FT,PL} &= [10^{-7}, 10^{-5}] \\
a_{PL,FT} &= [-10^{-5}, -10^{-7}] \\
a_{PL,DP} &= [10^{-7}, 10^{-5}] \\
a_{PL,HM} &= [10^{-4}, 10^{-2}] \\
a_{PL,PLe} &= [10^{-4}, 10^{-2}] \\
a_{PL,CM} &= [10^{-4}, 10^{-2}] \\
a_{DP,PL} &= [-10^{-5}, -10^{-7}] \\
a_{DP,HM} &= [10^{-4}, 10^{-2}] \\
a_{HM,PL} &= [10^{-4}, 10^{-2}] \\
a_{HM,DP} &= [10^{-4}, 10^{-2}] \\
a_{PLe,PL} &= [10^{-4}, -10^{-2}] \\
a_{CM,PL} &= [10^{-4}, -10^{-2}]
\end{aligned}$$

Article references

- **Chapter 2:**
García-Callejas, D., Molowny-Horas, R., & Araújo, M. B. (2018). Multiple interactions networks: towards more realistic descriptions of the web of life. *Oikos*, 127, 5-22. doi: 10.1111/oik.04428
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- **Chapter 4:**
García-Callejas, D. On the variability of Species Abundance Distributions with trophic guild and community structure. Under review in *Global Ecology and Biogeography*
- **Chapter 5:**
García-Callejas, D., Gravel, D., Molowny-Horas, R., & Araújo, M. B. Manuscript in preparation.
- **Chapter 6:**
García-Callejas, D., Molowny-Horas, R., & Araújo, M. B. Manuscript in preparation.

N.B.: the code used to generate the results of the different chapters is on the process of being cleaned up, and will be uploaded in its entirety to a public repository, at <https://github.com/DavidGarciaCallejas/>. As of 20/09/2018, most of the functions related to the model of chapter 3, and the full R code for replicating the results of chapters 4, 5, and 6, are already available. Likewise, the latex files for generating this document are also available. The illustrations for the heading pages of the chapters are all in the public domain, and the authors/name of the works are:

- Introduction: Haboku Sansui, by Sesshū Tōyō
- Chapter 2: El Perro, by Francisco de Goya
- Chapter 3: Street art by El Niño de las Pinturas
- Chapter 4: Street art by El Niño de las Pinturas
- Chapter 5: Crow and the Moon, by Kawanabe Kyōsai
- Chapter 6: Photograph of Cerro Huenes, Granada, taken by the author
- General Discussion: Photograph of a road in Nicaragua, taken by the author

COLOPHON

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